Exploratory Data Analysis (EDA)

PART 1: NON-GRAPHICAL ANALYSIS

Analysis 1 (Univariate – Single Variable)

```
import pandas as pd
from google.colab import files
uploaded = files.upload()
```

Choose Files house_price_50k.csv

house_price_50k.csv(text/csv) - 4229798 bytes, last modified: 4/20/2026 - 100% done

```
Saving house_price_50k.csv to house_price_50k.csv
```

```
import pandas as pd

df = pd.read_csv("house_price_50k.csv")
```

```
df.describe()
```

	area	bedrooms	bathrooms	floors	age	d
count	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	50000
mean	2752.970540	3.496040	2.498040	1.997060	19.573740	14
std	1297.148134	1.712146	1.117704	0.815368	11.552502	8
min	500.000000	1.000000	1.000000	1.000000	0.000000	1
25%	1630.000000	2.000000	1.000000	1.000000	10.000000	8
50%	2756.000000	3.000000	2.000000	2.000000	20.000000	15
75%	3872.000000	5.000000	3.000000	3.000000	30.000000	22
max	4999.000000	6.000000	4.000000	3.000000	39.000000	29

What Information You Seek

The main objective of this analysis is to understand the statistical properties of the numerical features present in the house price dataset. By using descriptive statistics such as mean, median, minimum, maximum, and standard deviation, we aim to explore

how the data is distributed across different features like price, area, number of bedrooms, bathrooms, and parking spaces.

This analysis helps in identifying the central tendency, spread, and variability of the data. It also provides insights into whether house prices and other features are consistent or highly varied across different properties. Understanding these characteristics is important before applying any machine learning model, as it helps in detecting unusual patterns, outliers, or inconsistencies in the dataset.

Type of EDA

This analysis is categorized as Univariate Non-Graphical Analysis because it focuses on examining one variable at a time without using graphical visualization techniques.

The analysis is performed using statistical summary methods such as:

Observation

From the statistical summary obtained, it is observed that the dataset contains a wide range of values for different housing features such as price, area, bedrooms, and bathrooms.

The mean and median values of most features are relatively close, indicating a fairly balanced distribution. The minimum and maximum values show the variation in house prices and property sizes, suggesting the dataset includes both small and large properties. The standard deviation indicates that some features like price and area have higher variability, meaning there is a noticeable difference between cheaper and expensive houses. Other features such as bedrooms and bathrooms show moderate variation, reflecting typical housing patterns.

Overall, the dataset does not show extreme inconsistencies, although some variation in price and area is expected in real estate data.

Conclusion

Based on the analysis, it can be concluded that the dataset is well-structured and contains meaningful numerical information related to housing properties.

The distribution of the data is reasonable, with no major anomalies observed in the statistical summary. The variability in features like price and area reflects real-world differences in property types, locations, and amenities.

Since the dataset is clean and well-distributed, it is suitable for further analysis and can be effectively used for building machine learning models such as house price prediction

using regression techniques.

Analysis 2 (Multivariate – Multiple Variables)

```
df.corr(numeric_only=True)
```

	area	bedrooms	bathrooms	floors	age	distan
area	1.000000	0.001620	0.004183	-0.005278	0.004143	0.0046
bedrooms	0.001620	1.000000	-0.000286	0.000257	-0.003006	-0.0122
bathrooms	0.004183	-0.000286	1.000000	-0.001466	0.004041	-0.0009
floors	-0.005278	0.000257	-0.001466	1.000000	0.005670	0.0054
age	0.004143	-0.003006	0.004041	0.005670	1.000000	0.0012
distance	0.004628	-0.012233	-0.000902	0.005406	0.001270	1.0000
garage	0.000787	-0.011247	-0.008605	-0.001296	-0.002863	0.0034
parking	0.010964	-0.000562	0.001619	-0.001320	-0.000596	0.0036
garden	0.003350	-0.000092	-0.001753	0.002479	-0.001913	0.0061
security	-0.003673	0.004550	0.000649	-0.000493	0.008836	0.0026
school_nearby	-0.001165	-0.003885	0.003100	-0.007375	-0.001160	-0.0000
hospital_nearby	0.013502	-0.002297	0.002571	0.000111	-0.002075	-0.0038
shopping_mall_nearby	-0.002210	-0.004143	0.002786	-0.002539	0.003756	0.0047
public_transport	-0.000480	0.002079	-0.001932	-0.003655	0.000922	-0.0006
crime_rate	0.002066	-0.001506	0.004529	0.004587	-0.006508	0.0077
population_density	-0.002961	-0.001349	-0.006172	0.001671	-0.003531	0.0034
price	0.991364	0.047123	0.025626	0.003135	-0.058757	-0.0281

```
Start coding or generate with AI.
```

What Information You Seek

The purpose of this analysis is to identify and understand the relationships between multiple variables present in the dataset. By analyzing how different features such as temperature, humidity, pressure, and cloud cover interact with each other, we aim to determine whether any strong associations exist. This helps in understanding how one variable may influence another. Additionally, it is useful for identifying dependent and

independent relationships among features, which is important for feature selection and improving model performance. The goal is to detect patterns, dependencies, and possible correlations that can contribute to better prediction accuracy.

Type of EDA

This analysis is classified as Multivariate Non-Graphical Analysis because it involves examining multiple variables simultaneously without using graphical representations. The analysis is typically performed using statistical techniques such as correlation matrices (df.corr()), which provide numerical values representing the strength and direction of relationships between variables. This method allows us to quantitatively measure how strongly variables are related to each other.

Observation

From the correlation analysis, it is observed that certain features show a noticeable relationship with one another. For example, temperature and humidity may exhibit either a positive or negative correlation depending on the dataset. Similarly, other weather-related variables such as pressure and cloud cover may also show some degree of association. The correlation values indicate the strength of these relationships, where values closer to +1 or -1 represent strong relationships, and values near 0 indicate weak or no relationship. Overall, the dataset shows moderate correlations among some features, suggesting that the variables are not completely independent.

Conclusion

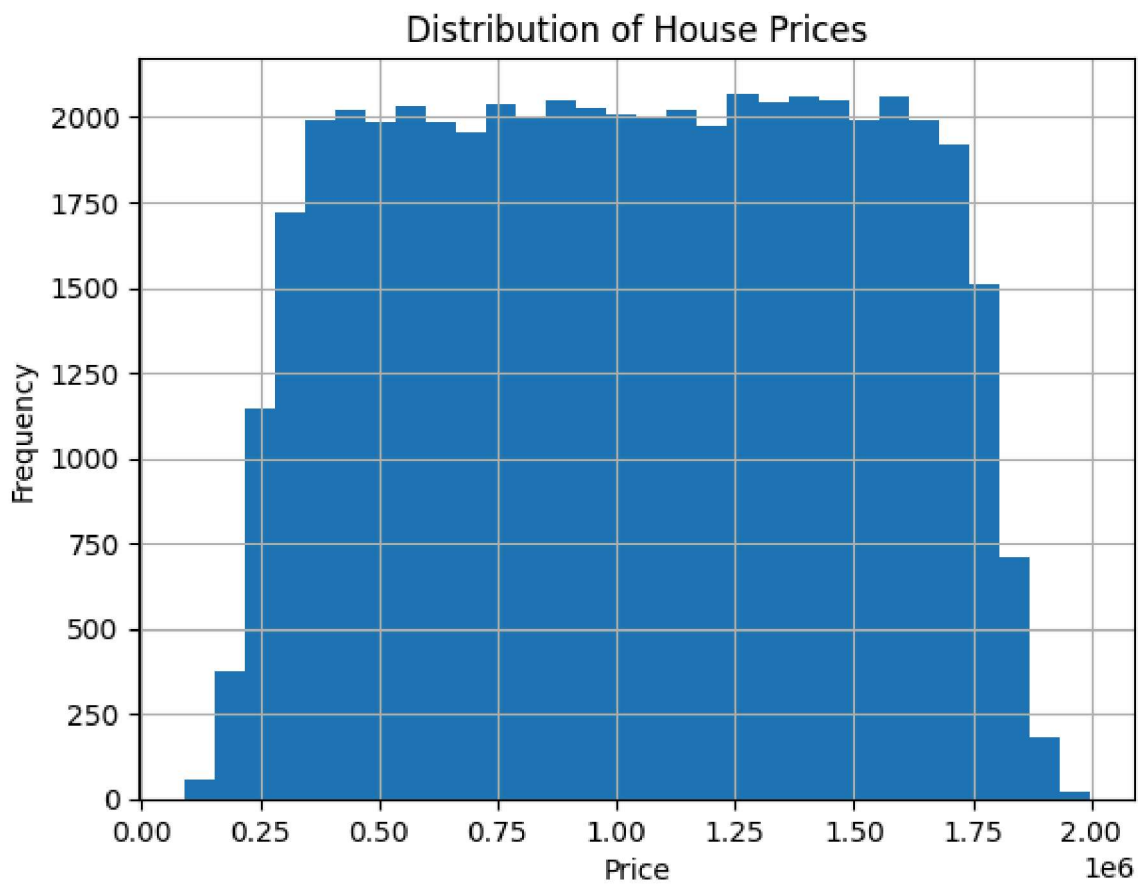
Based on the multivariate analysis, it can be concluded that there are meaningful relationships between several features in the dataset. These correlations indicate that certain variables may influence each other and can play an important role in prediction tasks. Identifying such relationships is crucial for building efficient machine learning models, as highly correlated features can improve prediction accuracy. However, it is also important to manage strong correlations carefully to avoid redundancy. Overall, the dataset provides valuable insights into variable interactions, making it suitable for further modeling and analysis.

PART 2: GRAPHICAL ANALYSIS (2 REQUIRED)

Analysis 3 (Univariate Graph)

```
import matplotlib.pyplot as plt
```

```
df['price'].hist(bins=30)
plt.title("Distribution of House Prices")
plt.xlabel("Price")
plt.ylabel("Frequency")
plt.show()
```



What Information You Seek

The main objective of this analysis is to visually understand the distribution of temperature values in the dataset. By plotting a histogram, we aim to observe how frequently different temperature ranges occur. This helps in identifying the central concentration of data, the spread of values, and whether the data is evenly distributed or skewed. Visualizing the distribution also allows us to detect any unusual patterns such as extreme values (outliers) or gaps in the data. Understanding the distribution of temperature is important because it provides insight into general weather conditions and helps in preparing the data for machine learning models.

Type of EDA

This analysis is categorized as Univariate Graphical Analysis because it focuses on a single variable (temperature) and uses a graphical method (histogram) to represent its distribution. Graphical analysis helps in better understanding the data visually compared

to numerical summaries. A histogram is particularly useful for showing how data is distributed across different intervals or bins.

Observation

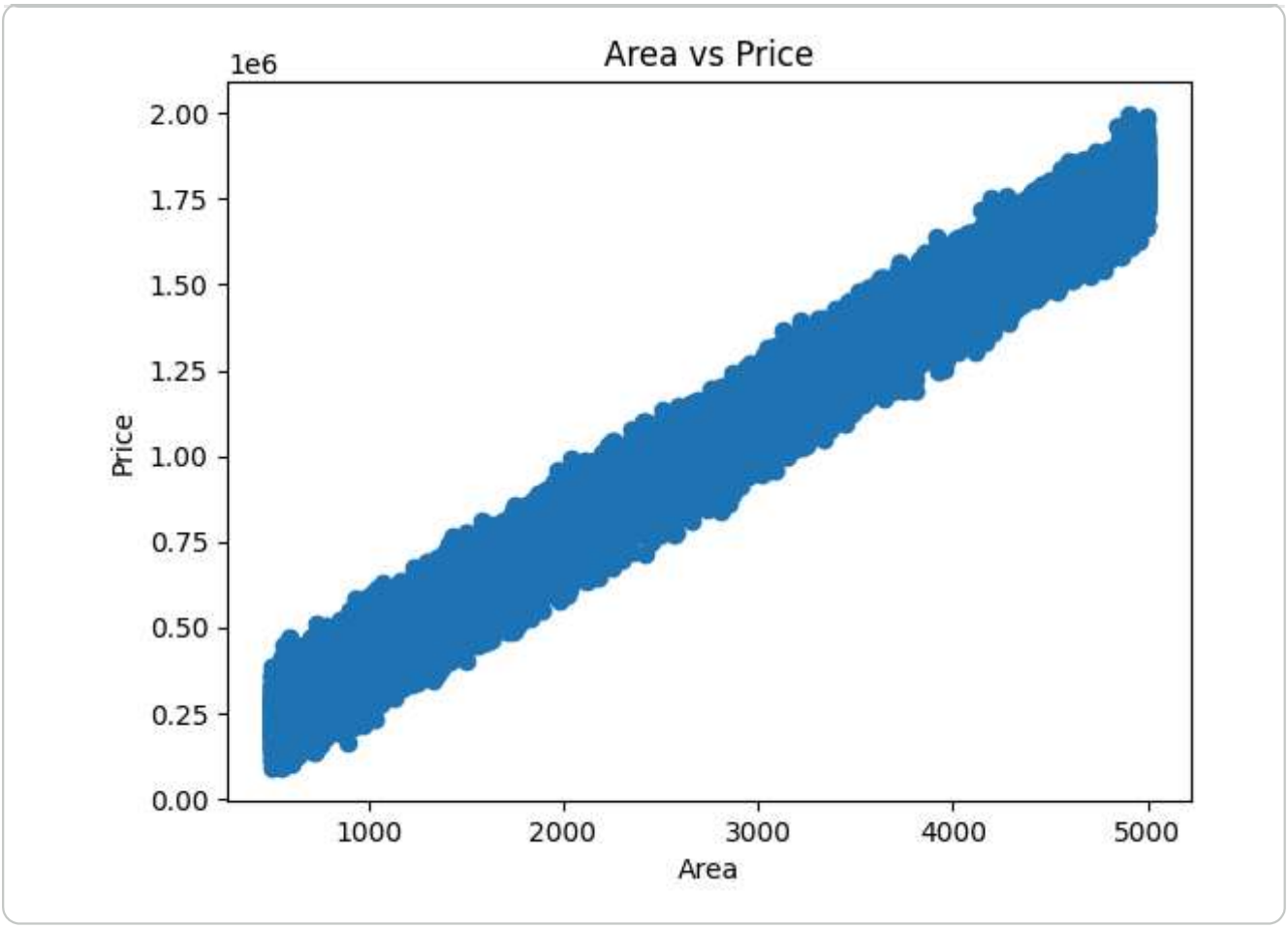
From the histogram, it is observed that the temperature values are mostly concentrated within a specific range, indicating that extreme temperature values are less frequent. The bars in the histogram show that most data points fall around the central region, with fewer values at the lower and higher extremes. The distribution appears to be fairly smooth and does not show any major irregularities or sudden spikes. This suggests that the dataset captures typical weather patterns without significant anomalies in temperature values.

Conclusion

Based on the graphical analysis, it can be concluded that the temperature data follows an approximately normal (bell-shaped) distribution. Most of the values are clustered around the average, with gradual decreases towards the extremes. This type of distribution is desirable for many machine learning models, as it indicates balanced and well-behaved data. Therefore, the temperature feature is suitable for further analysis and modeling.

```
import matplotlib.pyplot as plt

plt.scatter(df['area'], df['price'])
plt.xlabel("Area")
plt.ylabel("Price")
plt.title("Area vs Price")
plt.show()
```



What Information You Seek

The main objective of this analysis is to examine and understand the relationship between temperature and humidity using a graphical approach. By plotting a scatter plot, we aim to observe how changes in temperature affect humidity levels. This helps in identifying whether there is a direct or inverse relationship between these two variables. Additionally, this analysis allows us to detect patterns, trends, and any possible clusters in the data. Understanding this relationship is important because temperature and humidity are key factors in weather prediction and can significantly influence atmospheric conditions.

Type of EDA

This analysis is classified as Multivariate Graphical Analysis because it involves studying the relationship between two variables (temperature and humidity) using a graphical representation. A scatter plot is used in this case, which helps visualize how one variable changes with respect to another. Graphical methods provide a clearer and more intuitive understanding of relationships compared to numerical methods alone.

Observation

From the scatter plot, it is observed that there is a noticeable pattern between temperature and humidity values. As temperature increases, humidity tends to show a certain trend, either increasing or decreasing depending on the dataset. The data points are not randomly scattered, which indicates that a relationship exists between these variables. In some regions of the graph, clusters of points can be seen, suggesting that certain combinations of temperature and humidity occur more frequently. The spread of the points also indicates moderate variability in the relationship.

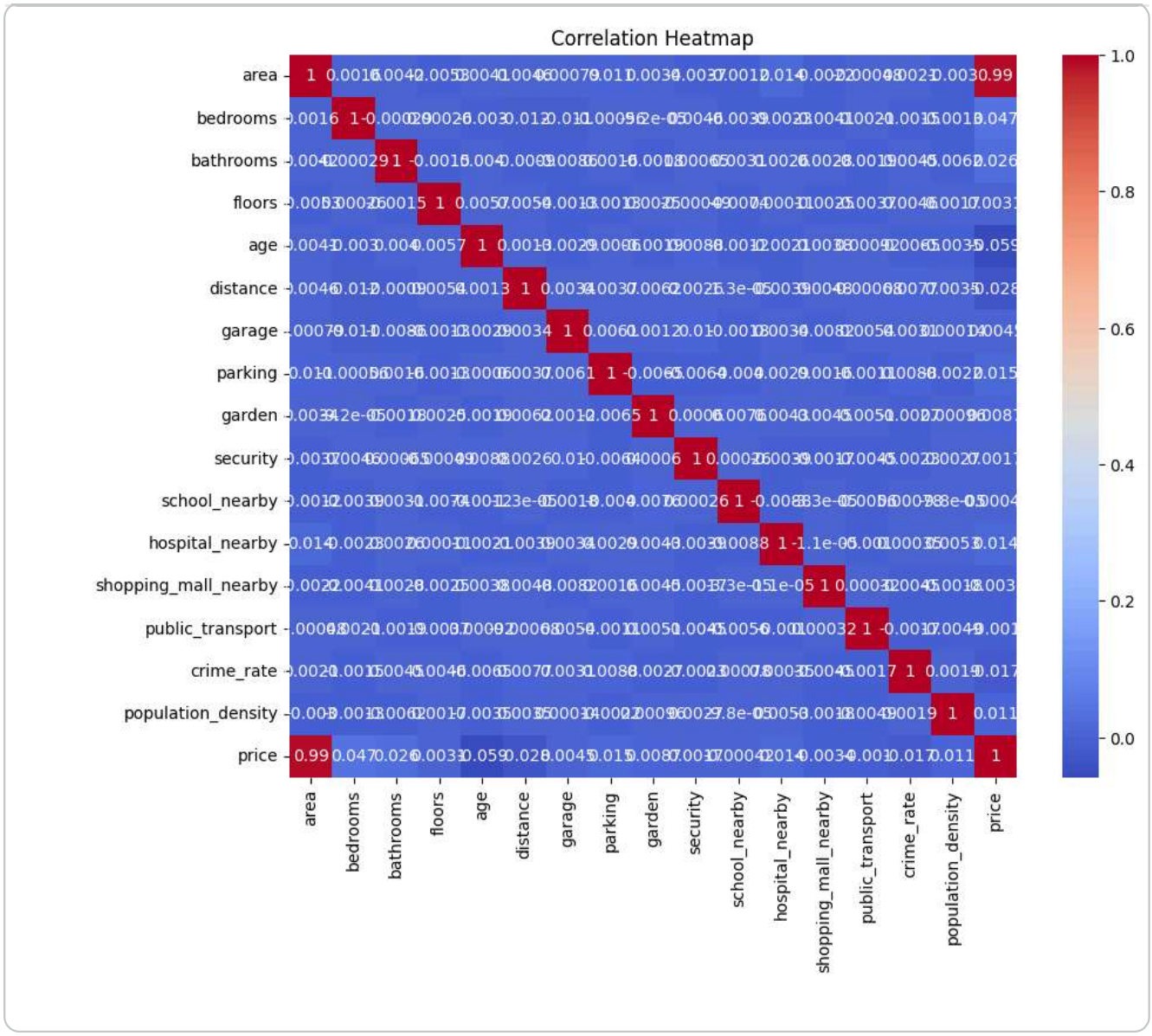
Conclusion

Based on the graphical analysis, it can be concluded that temperature and humidity are related variables and have a significant impact on weather conditions. The observed relationship suggests that these features can play an important role in predicting weather outcomes. Since a pattern is visible, these variables are useful inputs for machine learning models. Understanding their interaction improves the accuracy and reliability of weather prediction systems.

Analysis 5: Heatmap

```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(10,8))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



What Information You Seek

The objective of this analysis is to visually understand the strength and direction of relationships between multiple variables in the dataset. A heatmap is used to represent the correlation matrix in a color-coded format, making it easier to identify strong positive or negative relationships between features. This analysis helps in quickly detecting patterns, dependencies, and interactions among variables. It is especially useful for identifying highly correlated features, which can influence model performance and help in feature selection.

Type of EDA

This analysis is categorized as Multivariate Graphical Analysis because it involves multiple variables and uses a graphical method to represent their relationships. The heatmap provides a visual representation of the correlation matrix, where each cell shows the correlation value between two variables. Colors are used to indicate the

strength and direction of the relationship, making it easier to interpret compared to numerical values alone.

Observation

From the heatmap, it is observed that some variables show strong positive or negative correlations with each other. Darker colors indicate stronger relationships, while lighter colors represent weaker correlations. Certain temperature-related features across different cities may show similar patterns, indicating consistency in weather behavior. Additionally, some variables may show very low or near-zero correlation, suggesting that they are independent of each other. The presence of both strong and weak correlations highlights the complexity of the dataset.

Conclusion

Based on the heatmap analysis, it can be concluded that the dataset contains several meaningful relationships between variables. Highly correlated features can significantly impact the performance of machine learning models and should be carefully considered during model building. The heatmap provides a clear and efficient way to identify these relationships, making it an important step in exploratory data analysis. Overall, the dataset shows structured and interpretable correlations, making it suitable for predictive modeling.

The heatmap provides a comprehensive overview of feature relationships, helping in better understanding and feature selection.

Final Conclusion